# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **AI-enabled enterprise workflows**, **microservice-style backends**, and **full-stack web applications** using TypeScript, Python, Java, SQL, cloud services, and CI/CD automation.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

## Experience

**USC Dornsife** **LA, California**
*Software Engineer* *September 2025 - Present*
- Built **multi-tenant backend systems** for 500+ users using **Azure SQL/PostgreSQL**, reducing latency from **1.2s to 350ms** through persistence-layer refactoring and query-path optimization.
- Implemented **AI-driven validation workflows** for 5K+ monthly jobs with FastAPI orchestration, improving reliability, deployment safety, and maintainability for rapidly evolving data-heavy product features.

**Grad8** **LA, California**
*Software Engineer Intern* *June 2025 - August 2025*
- Developed **full-stack features** with Flask + React handling 10K+ daily requests, translating product requirements into secure APIs, scalable services, and user-facing interfaces in Agile delivery cycles.
- Added **automated CI/CD controls** and concurrent processing safeguards, reducing turnaround time by **60%** while improving release quality and predictability under strict delivery timelines.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 - July 2024*
- Engineered **API and ETL microservice workflows** using OOP, AWS Glue, and S3, improving response time by **30%** across distributed, cloud-based data integration paths.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 - August 2023*
- Built **MERN applications** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving collaboration workflows and operational reliability for production users.
- Automated deployment/testing with **Docker** and Jenkins, reducing deployment time by **50%** while strengthening release consistency and post-deployment quality across iterative feature rollouts.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 - May 2025*
- Developed **Python simulation pipelines** for 100K+ trajectory logs, enabling first-principles evaluation of model behavior and faster iteration across AI-assisted optimization experiments.
- Documented reproducible validation outputs and debugging findings, improving team communication and accelerating conversion of experimental prototypes into production-ready engineering components.

## Skills

**Languages**: **TypeScript**, **Python**, **Java**, SQL, JavaScript, C++
**Full Stack**: React.js, Next.js, Flask, FastAPI, Node.js, REST APIs, AngularJS/Dojo familiarity
**Cloud & Platform**: Microsoft **Azure** concepts, Kubernetes, Docker, AWS (S3/Glue/RDS), multi-tenant architecture
**Data & Persistence**: PostgreSQL, MySQL, SQL Server patterns, MongoDB, persistence-layer refactoring
**Delivery**: Git, GitHub, CI/CD automation, unit/integration testing, Scrum/Agile collaboration, AI-assisted development workflows

## Projects

**FinNexus** Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **scalable microservice-style workflows** for 50K+ users and 1M+ transactions with real-time updates, secure integrations, and deployment patterns aligned to enterprise collaboration use cases.

**Chat2SQL** Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed an **AI-assisted data platform** across 10+ databases with schema-aware query generation and execution validation, enabling faster analytics workflows and safer enterprise decision support.

**HMIS** Live | *MERN, ABDM APIs, Twilio*
- Designed **workflow-centric web services** supporting 100+ daily appointments, integrating API-driven automation and monitoring checks that improved record retrieval by **2x** and reduced operational friction.